

## **Statistical Analysis of Self-Reported Academic Data: Limitations and Lessons Learned from a Case Study of Grad Café**

The empirical constraints of using anonymously submitted, self-reported data from platforms like Grad Café for statistical analysis greatly limit the generalizability of the findings to real-world situations. The main concern arises from non-random sampling, which causes severe selection bias and self-reporting bias. Grad Café users are by nature a self-selected group, skewed toward highly motivated, digitally engaged applicants, particularly in quantitative fields such as Computer Science and STEM. Thus, the dataset is entirely unrepresentative of the broader graduate applicant population. Applicants with extreme results, such as early acceptances to top-tier schools or very high standardized test scores, are far more likely to post their results voluntarily. Furthermore, the presence of missing values, inconsistent formatting and unverified data entries degrades the statistical fidelity of the metrics. So, while our database calculates certain descriptive statistics mathematically, these are numbers for a non-random, self-selected subset and not the objective reality of the global applicant pool.

Our empirical analysis of standardized test metrics illustrates this distinction between mathematical computation within the database and valid real-world inference. We found the average GRE Quantitative score in our synthetic data to be about 162.50, significantly above the broader reference population average, which is closer to 157 for all test-takers. This discrepancy does not mean that the worldwide applicant pool has a mean of 162.50. Instead, it points to a systemic bias of non-response. Those who have high quantitative scores are much more likely to submit their scores on their own accord to demonstrate their competitive advantage, while those who have average or below average scores often leave these optional scores blank altogether. Similarly, our analysis indicated an International student representation of 13.28% and an overall Fall 2025 acceptance rate of 46.67%, both of which likely reflect posting biases—where accepted candidates eager to share good news and specific international applicant cohorts are overrepresented—rather than true institutional admission rates. In the end, researchers must view this database as an observational sample of self-selected user activity, not a ground-truth representation of graduate admissions.